

SESSION-04
SECURITIES ANALYSIS & PORTFOLIO MANAGEMENT

From Return Measurement to Portfolio Pricing

A Capital Market Theory Journey



SESSION 1

What did I earn?

Return & risk measurement



SESSION 2

How do assets interact?

Covariance, correlation, beta, alpha



SESSION 3

What's the best combination?

Portfolio risk-return & diversification



SESSION 4

Where's the efficient boundary?

Efficient frontier, MVP, CML



SESSION 5

What should I be paid for risk?

CAPM & SML



SESSION 6

Did my manager add value?

Risk-adjusted performance evaluation

"By the end of these 6 sessions, you should be able to open any mutual fund fact sheet and know exactly which of these six questions its marketing is quietly avoiding. Which one would you guess — and why?"

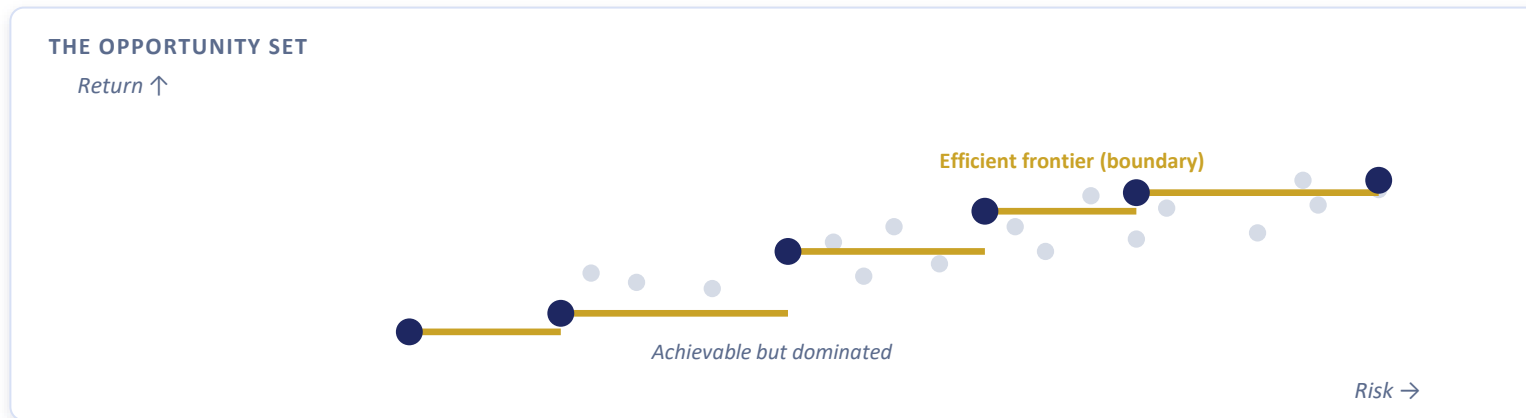


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The Portfolio Opportunity Set: Mapping All Feasible Combinations

- The opportunity set: every possible risk-return combination achievable from a given set of assets
- The efficient frontier is its upper-left boundary — not the whole set
- Every interior point is achievable but dominated by some point on the boundary



WORKED EXAMPLE

Equal-weight, two uncorrelated risky assets — final $E(R_p)$ and σ_p only, no arithmetic shown.

REFLECTION POINT

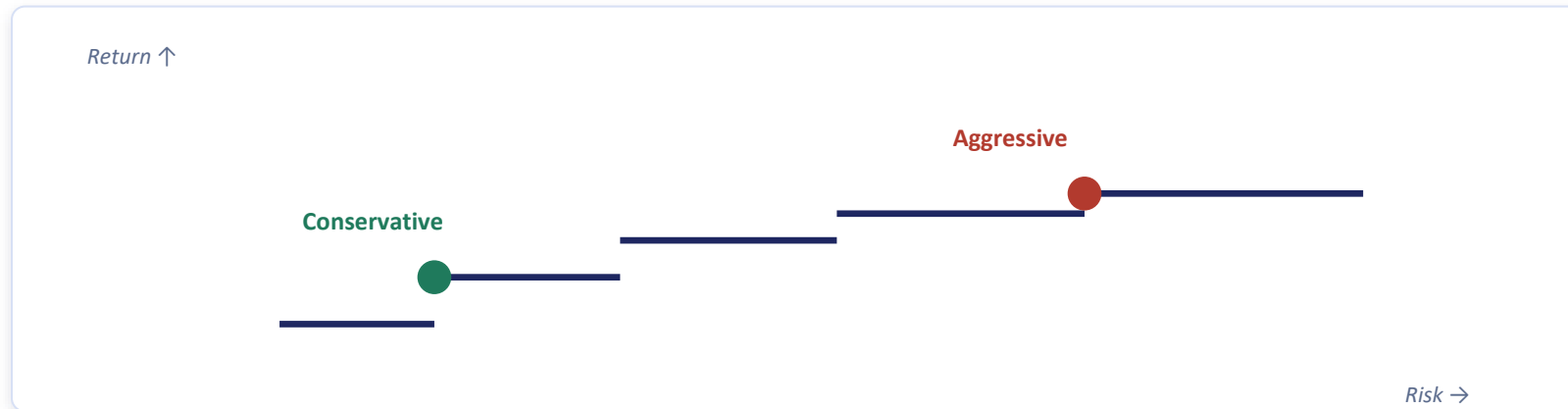
“Every dot inside the boundary is still a real, investable portfolio. Why would a rational investor never choose one?”

QUESTION BANK LINK

Use in class: Problem 5.1 (equal investment, two uncorrelated risky assets)

The Efficient Frontier: The Geometry of Rational Investing

- Efficient portfolio: highest expected return for a given risk level (or lowest risk for a given return)
- Rational investors never knowingly hold a dominated (inefficient) portfolio
- The frontier curve is a decision rule, not just a picture — dominance is a testable condition



Conservative

Low correlation allocation

Aggressive

Negative correlation allocation

Both efficient — neither is “more correct”

REFLECTION POINT

“Two portfolios both sit exactly on the efficient frontier — one conservative, one aggressive. Is one of them ‘more correct’ than the other?”

QUESTION BANK LINK

Use in class: Problem 5.2 (conservative allocation, low correlation)

Assign for practice: Problem 5.3 (aggressive allocation, negative correlation)

Dominated vs Efficient Portfolios: Which Portfolio Deserves Capital?

- Alpha dominates Beta (higher return, lower risk) and dominates Gamma (same return, lower risk)
- Historical optimization does not guarantee future efficiency — estimation risk can make a “backtested optimal” portfolio disappoint out-of-sample
- Past efficiency is not future efficiency

Portfolio	Return	Risk	Sharpe*	Verdict
Alpha	12%	18%	0.33	✓ Efficient
Beta	10%	20%	0.20	Dominated
Gamma	12%	22%	0.27	Dominated

**Illustrative Sharpe ratios at $R_f=6\%$. Alpha beats Beta outright and matches Gamma's return at lower risk — dominance in both directions.*

REFLECTION POINT

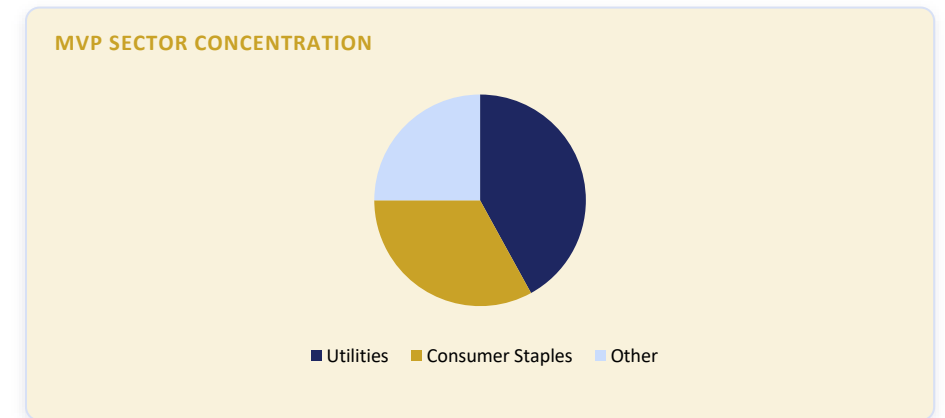
“If Alpha dominates both Beta and Gamma today, based on historical data, how confident should you be that Alpha will still dominate them next year?”

QUESTION BANK LINK

None directly — this slide reinforces Problems 5.1–5.3 conceptually rather than introducing a new numerical problem.

The Minimum Variance Portfolio: Safe or Dangerous?

- MVP: the leftmost point on the efficient frontier — the lowest achievable variance from a given asset set
- MVP optimization often produces extreme sector concentration — trading idiosyncratic stock risk for sector-concentration risk
- Regime-dependent: defensive in calm markets, but “defensive” correlations can collapse in a structural crisis (2008)



REFLECTION POINT

“If MVP investing quietly concentrates a portfolio into a handful of defensive sectors, is ‘minimum variance’ actually a form of diversification — or a disguised form of concentration?”

QUESTION BANK LINK

Use in class: Problem 9.7 or 9.9 (MVP weights/risk; MVP vs equal-weighted)

Assign for practice: Problem 9.8 (zero-risk case, perfect negative correlation)

Estimation Risk: The Hidden Enemy of Optimization

- Optimizers amplify small input errors — a portfolio that looks optimal on paper can underperform badly out-of-sample
- Practitioner responses: the Single-Index Model reduces the burden to n beta estimates; multi-factor models (Fama-French, Barra, Black-Litterman) impose further structure
- Overengineered optimization frequently underperforms a simple equal-weight or 60/40 portfolio

Securities	Covariance Pairs Needed
10	45
100	4,950
500	124,750

GARBAGE IN, GARBAGE OUT

Every one of these thousands of pairwise estimates carries some error. Optimizers don't average that error away — they amplify it into portfolios that look perfect on paper and disappoint out-of-sample.

REFLECTION POINT

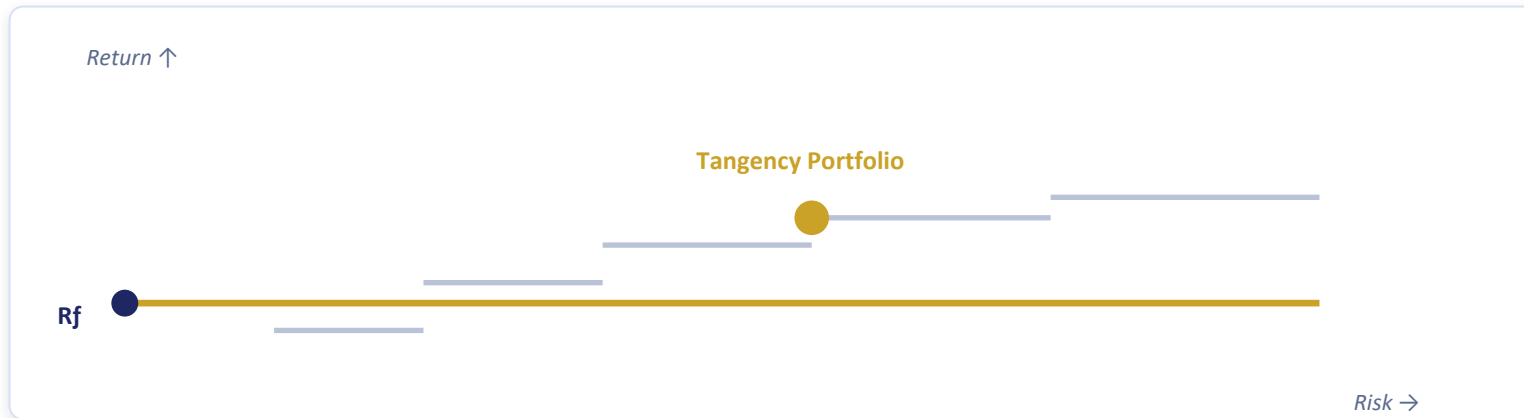
“If a 100-security optimization needs 4,950 correlation estimates and each one carries even a small error, should institutional investors trust a fully optimized portfolio more, or less, than a simple 60/40 split?”

QUESTION BANK LINK

None — practitioner-caution slide, directly building on Session 3's Slide 29 covariance-equation scaling point.

Adding the Risk-Free Asset: A Straight Line Transforms the Problem

- Introducing a risk-free asset converts the curved efficient frontier into a straight line from R_f , tangent to the risky-asset frontier
- Every combination of R_f and the tangency portfolio dominates every point on the curved frontier except the tangency point itself
- Markowitz (risky-asset frontier) → Tobin (adds the risk-free asset) → Sharpe/Lintner (CAPM, Session 5)



THE HINGE POINT

Arguably the single most important geometric transition in the entire course.

REFLECTION POINT

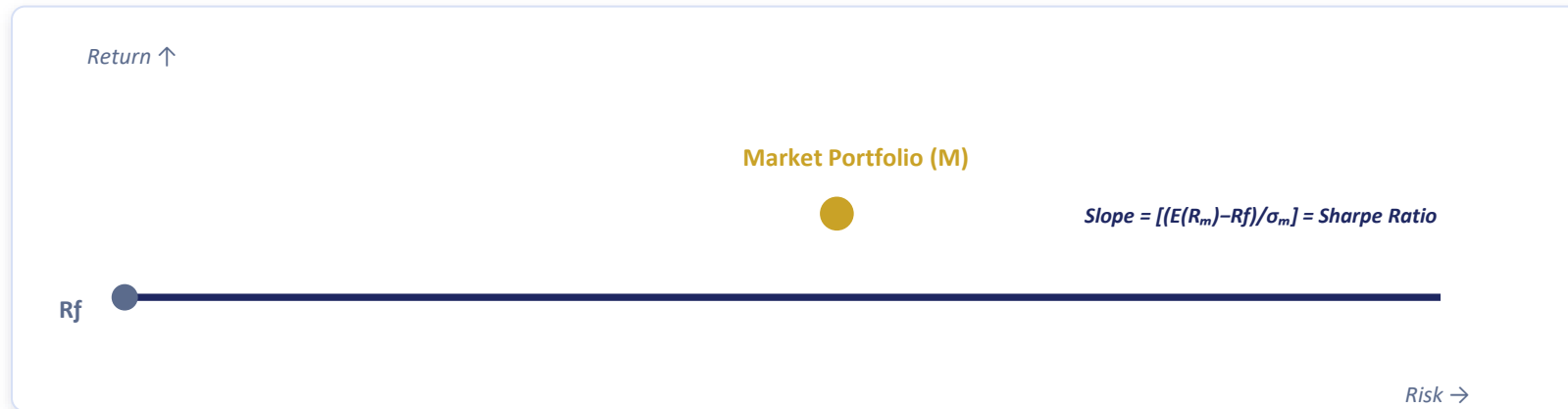
“Why is a straight line analytically more powerful than a curve here? What does 'combining' the risk-free asset with any risky portfolio actually let an investor do that a curve alone does not?”

QUESTION BANK LINK

None — conceptual bridge slide.

The Capital Market Line

- CML: $E(R_p) = R_f + [(E(R_m) - R_f) / \sigma_m] \times \sigma_p$ — reward per unit of total risk, anchored at the market portfolio
- The CML applies only to efficient (fully diversified) portfolios — not individual securities or inefficient portfolios
- The CML's slope is the Sharpe ratio — this reappears as the central performance metric in Session 6



SCOPE RESTRICTION

Efficient portfolios only. A single stock that isn't perfectly diversified cannot be priced by the CML — this is exactly what necessitates the SML in Session 5.

REFLECTION POINT

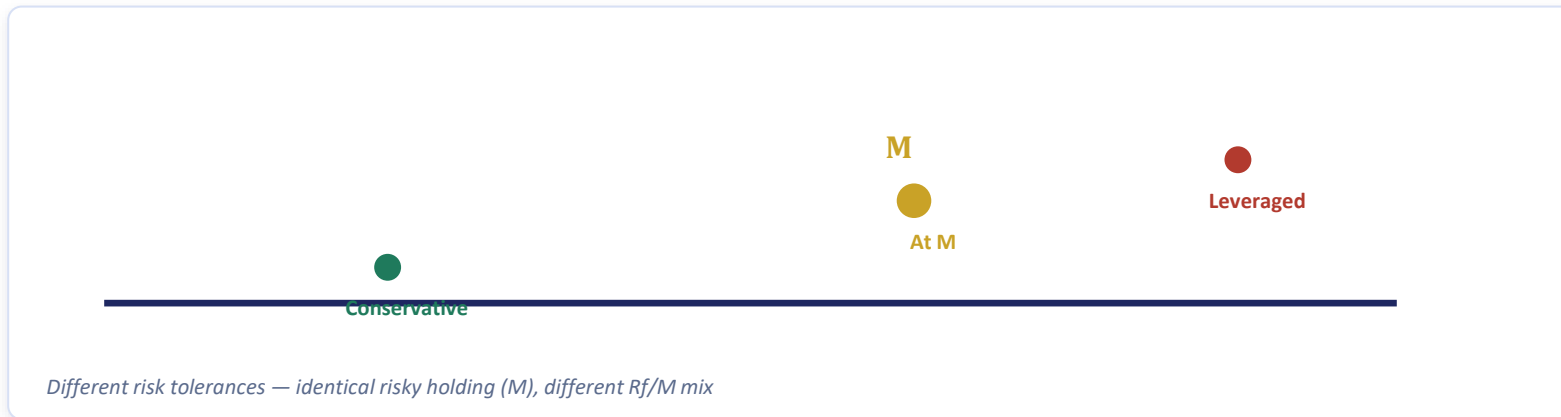
"The CML only prices efficient portfolios. What happens the moment you try to use it to price a single, individual stock that isn't perfectly diversified?"

QUESTION BANK LINK

Use in class: Problem 6.5 (pure market-portfolio position — sits exactly at the tangency point)

The Market Portfolio and the Separation Theorem

- Market portfolio: the single optimal risky portfolio all investors hold in equilibrium, regardless of risk preference
- Separation Theorem: the investment decision (which risky portfolio) separates completely from the financing decision (how much R_f vs risky exposure)
- In practice, no true “market portfolio” is observable — index funds are only a domestic-equity proxy (previewed as Roll's Critique, Session 5)



SEPARATION THEOREM

*Which risky portfolio?
(investment decision)*

*How much R_f vs risky?
(financing decision)*

Completely independent.

REFLECTION POINT

“If every rational investor is supposed to hold the exact same risky portfolio under CAPM, why does virtually no real investor’s actual portfolio look anything like a global market-cap-weighted index?”

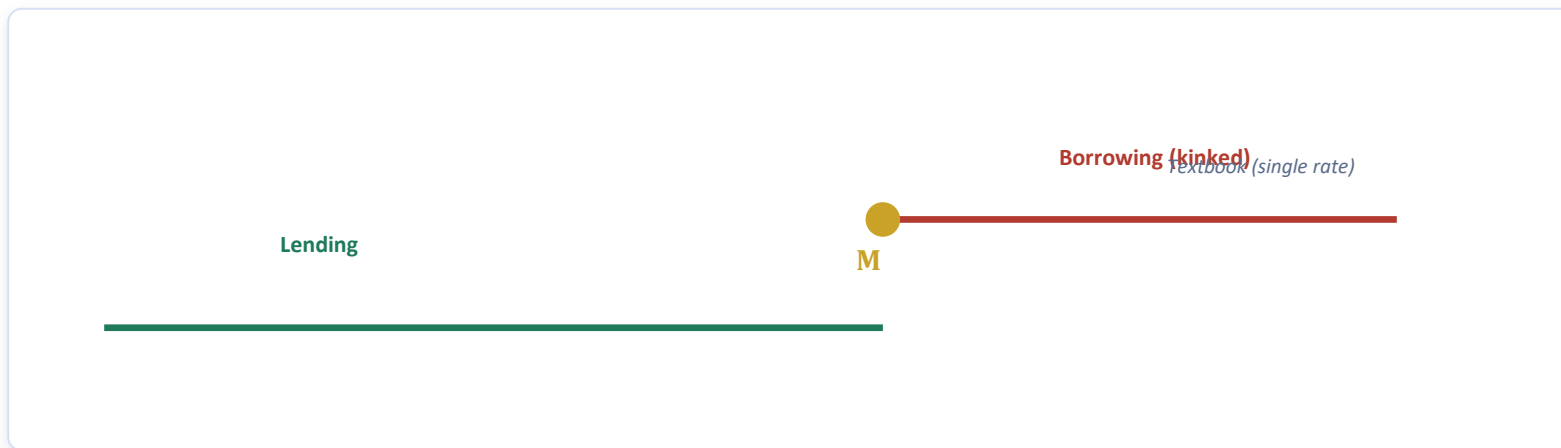
QUESTION BANK LINK

Use in class: Problem 9.10 (Separation Theorem, two investors/one M-proportion)

Assign for practice: Problems 9.11 (Roll's Critique in numbers), 9.12 (wealth-management implication)

Lending, Borrowing, and Leverage on the CML

- Three investor types along the CML: lending (partly R_f , partly market) | pure market portfolio (100% market) | borrowing/leveraged (>100% market, financed at R_f)
- The kinked CML: in reality, the borrowing rate exceeds the lending rate — this flattens the CML beyond the market portfolio



Conservative

Lending — part R_f , part M

Aggressive

Leveraged — borrows to hold >100% M

REFLECTION POINT

“If borrowing costs more than lending pays, does that make leveraged CML positions irrational, or simply less attractive than the textbook version — and where exactly is that line?”

QUESTION BANK LINK

Use in class: Problems 6.1 (conservative/lending), 6.3 (aggressive/leveraged)

Assign for practice: Problems 6.2, 6.4, 6.6, 6.7, 6.8 (kinked CML — advanced/challenge)

Session 4 Synthesis: From Portfolio Theory to Asset Pricing

- Toolkit now complete: opportunity set → efficient frontier → MVP → risk-free asset → CML → market portfolio/separation theorem → lending/borrowing/leverage
- The CML prices efficient portfolios only — individual, imperfectly diversified securities remain unpriced by anything covered so far



You are here → standing at the threshold of Sharpe/Lintner, about to cross into Session 5.

REFLECTION POINT

“The CML prices efficient portfolios. How would you go about pricing a single security that isn't perfectly diversified on its own?”

QUESTION BANK LINK

None — pure bridge/synthesis slide closing Session 4.